# Fundamental Physics 1: Mechanics

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Ngày 5 tháng 11 năm 2024

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# Chapter 2: Forces and Newton's laws of motion (1)

## Forces

- A force is a push or a pull
- Force is a vector quantity. It has both magnitude and direction
- The unit of force is newton (N)
- In general, a body can have several forces acting in it at the same time.

## Balanced forces and Newton's first law of motion

- No resultant external forces means that the external forces on the body are balanced.
- If the motion of a given body is not disturbed by external influences then that body moves with constant velocity
- $\vec{r} = \vec{r}_0 + \vec{v}.t, \quad \vec{r}_0 \quad , \quad \vec{v}$ are constant vectors
- if $\vec{v} \neq 0$ (the object is moving), object continues to move in a straight line with constant speed.
- If $\vec{v} = 0 \rightarrow$ the body simply remains at rest.

# Chapter 2: Forces and Newton's laws of motion (2)

## Unbalanced forces

- Forces acting on a body are unbalanced if the resultant force is not zero.
- Unbalanced forces can change the motion of an object. Unbalanced forces can cause:
  - a stationary object to move
  - the speed of an object to increase
  - the speed of an object to decrease
  - the direction of motion to change.

## Newton's second law of motion

- Momentum: $\vec{p} = m.\vec{v}$
- Second law:
$$\frac{d\vec{p}}{dt} = \vec{F} \rightarrow \text{net influence/force}$$
- $m = const \Rightarrow \vec{F} = m.\vec{a}$

## Newton's second law of motion

*The rate of change of momentum of an object is directly proportional to the applied force and it occurs along the direction of the force.*

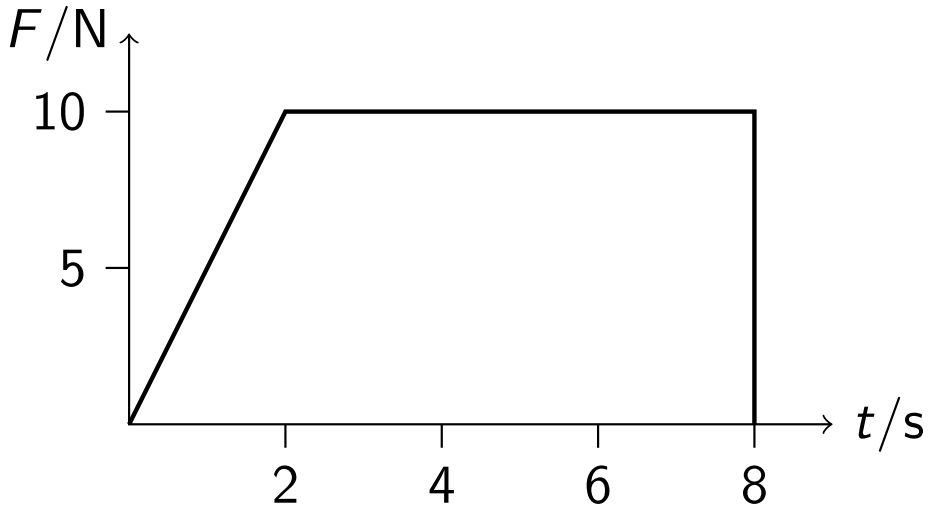
$$F = \frac{d}{dt}(mv)$$

$$\int_{t_1}^{t_2} F.dt = mv - mu$$

- $\int_{t_1}^{t_2} F.dt \Rightarrow$ *impulse* of the force $F \rightarrow$ the change in momentum.
- The impulse of a constant force is the product of the force and the time duration that it acts.
- *Impulse* of the force $F$ = area below $(F - t)$ graph between $t_1$ and $t_2$

# Chapter 2: Worked example

A body of mass 3 $kg$ experiences a force $F$ which varies with time $t$ as shown in the Figure below. What is the increase in momentum of the object after 8 $s$ ?



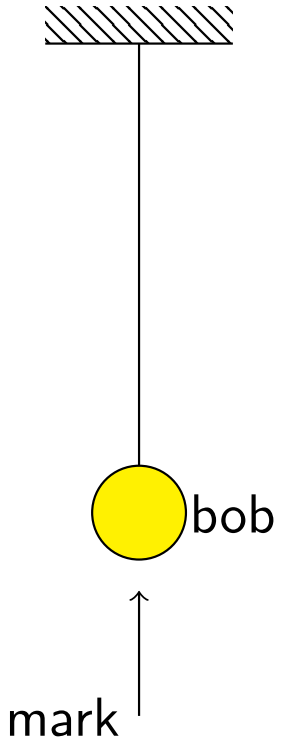
# Chapter 2: Worked example

A plumb-line is hung from the roof of a train carriage. The bob is vertically above a mark when the train is stationary. Draw figures to show the position of the bob relative to the mark when

(a) the train moves with uniform velocity,

(b) the train accelerates uniformly,

(c) the train retards uniformly.






## Newton's third law of motion

- $\vec{f}_{ab}$: force exerted on (a) by (b)
- $\vec{f}_{ab}$: force exerted on (b) by (a)
- Newton's third law: $\vec{f}_{ab} = -\vec{f}_{ba}$
- Why Newton's third law? $\rightarrow$ No self generated forces.

# Chapter 2: Everyday forces (1)

- **Gravitational force** (the force due to gravity):

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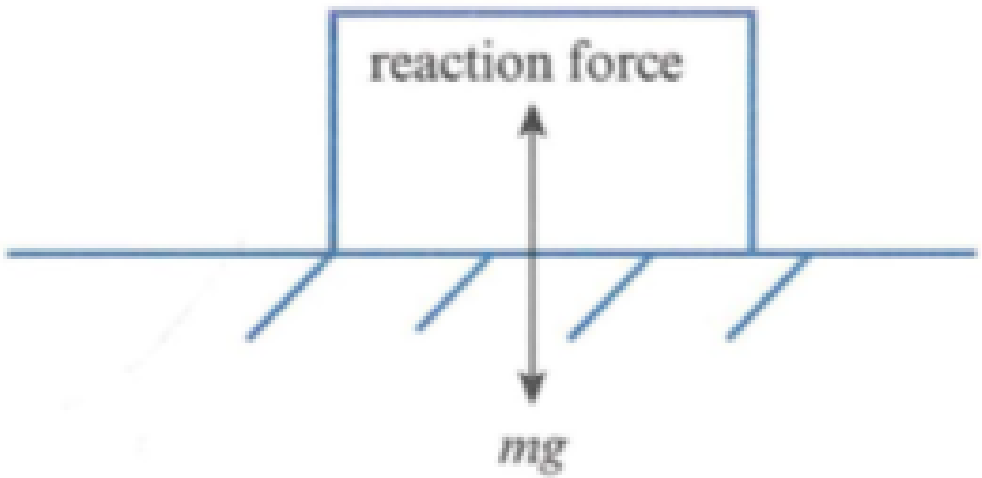
$$\vec{f_g} = m\vec{g}$$

- **Weight**: The weight of an object of mass $m$ kg is given by

$$W = mg$$

($g$ is free-fall acceleration).

- **Normal reaction/Reaction force**: The normal contact force between the object and the surface it is on is called the reaction force and is always perpendicular to the surface.



# Everyday forces

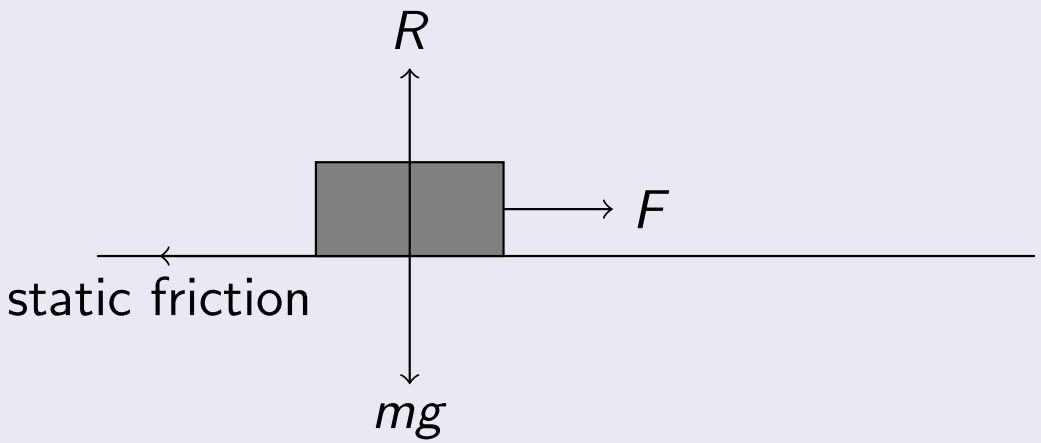
- Find normal reaction on an object resting on a table?

# Everyday forces

- Find normal reaction on an object of mass $m$ resting on the floor of an elevator moving up and down with acceleration $\vec{a}$?

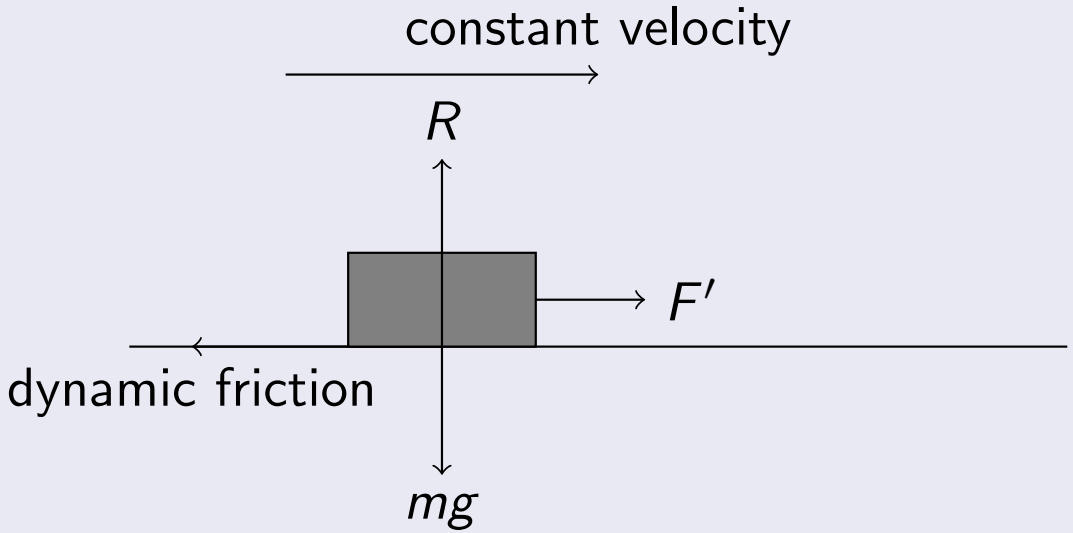
## Static friction

- The magnitude of the static friction is equal to applied force but opposite direction to the applied force.
- The maximum static friction occurs just before motion takes place. It is known as *limiting static friction*



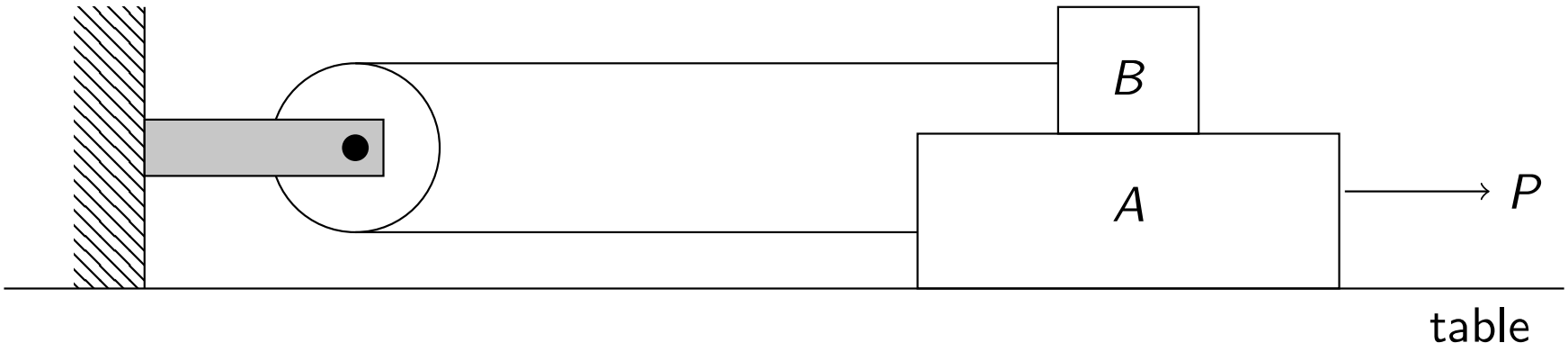
## Kinetic/dynamic friction

$f = \mu.R_n$, where $\mu$ : coefficient of friction and $R_n$ is normal reaction of the surface.



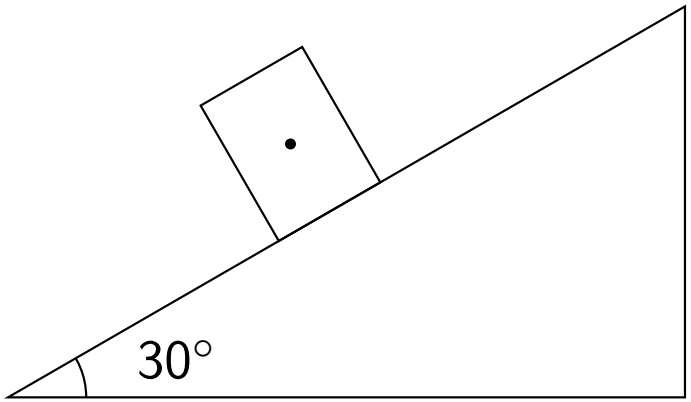
# Chapter 2: Worked example

The below figure shows a wooden block $B$ at rest on another block $A$. Both blocks are connected by a string which goes round a fixed smooth pulley. The dynamic friction between block $A$ and the table is 14 $N$ and that between block $B$ and $A$ is 10 $N$. What is the horizontal force $P$ which is required to move block $A$ with uniform velocity?

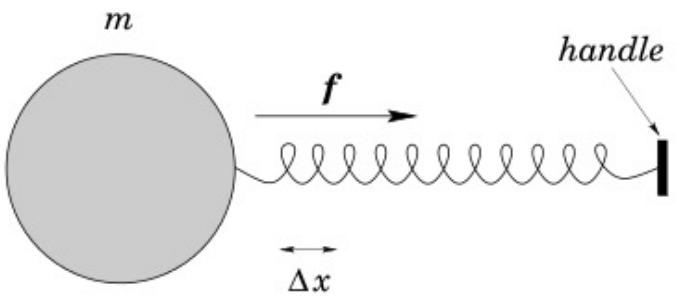


# Chapter 2: worked example

A metal block of mass 4 $kg$ is on a plane inclined at an angle of 30° to the horizontal. The block slides down the plane from rest and travels a distance of 1.0 $m$ in 2.0 $s$. What is the frictional force between the block and the inclined plane when the block is slipping?



# Chapter 2: Everyday forces: Hooke's law (3)




- The force exerted on an object by a coiled spring:

$$\vec{f} = -k.\Delta\vec{x}$$

  - $\Delta\vec{x} = \vec{l} - \vec{l_0} \rightarrow$ extension of the spring
  - $\vec{l} \rightarrow$ actual length; $\vec{l_0} \rightarrow$ natural length of the spring.
- Application of Hooke's law: to quantify the magnitude and direction of the force exerted on a given body by means of a spring
  - $F_0 = 1N = 1kg.m/s^2 = k.\Delta x_0$, where $\Delta x_0 \rightarrow$ "sample" extension
  - $F \propto \Delta x = \alpha.\Delta x_0 \Rightarrow \alpha$ is the value of the force we want to quantify.
  - Strain: $s = \Delta x / l_0$